

> restart;		
>		
> Q01 := 10	Q01 := 10	(1)
> Q02 := 1	Q02 := 1	(2)
> #Determine the firm that cannot reverse-engineer if Q01 > Q02 then xe1 := 0 end if; if Q02 > Q01 then xe2 := 0 end if;	xe1 := 0	(3)
> xe1 := 0; xe2 := 0;	xe1 := 0 xe2 := 0	(4)
> Dc := 1	Dc := 1	(5)
> Ec := 1	Ec := 1	(6)
> Mc := 100	Mc := 100	(7)
> L12 := 0.75	L12 := 0.7500	(8)
> P1 := S1 - Cd1 - Ce1	P1 := S1 - Cd1 - Ce1	(9)
> P2 := S2 - Cd2 - Ce2	P2 := S2 - Cd2 - Ce2	(10)
> $S1 := \frac{Mc \cdot Q1^2}{Q1 + L12 \cdot Q2}$	$S1 := \frac{100 Q1^2}{Q1 + 0.7500 Q2}$	(11)
> $S2 := \frac{Mc \cdot Q2^2}{L12 \cdot Q1 + Q2}$	$S2 := \frac{100 Q2^2}{0.7500 Q1 + Q2}$	(12)
> $Cd1 := Dc \cdot xd1 \cdot \left(1 + xe1 + \frac{xd1}{2}\right)$	$Cd1 := xd1 \left(1 + \frac{xd1}{2}\right)$	(13)
> $Cd2 := Dc \cdot xd2 \cdot \left(1 + xe2 + \frac{xd2}{2}\right)$		

$$Cd2 := xd2 \left(1 + \frac{xd2}{2} \right) \quad (14)$$

$$> Ce1 := \frac{Ec \cdot xe1}{L12}$$

$$Ce1 := 0. \quad (15)$$

$$> Ce2 := \frac{Ec \cdot xe2}{L12}$$

$$Ce2 := 0. \quad (16)$$

$$> Q1 := Q01 + xd1 + xe1$$

$$Q1 := 10 + xd1 \quad (17)$$

$$> Q2 := Q02 + xd2 + xe2$$

$$Q2 := 1 + xd2 \quad (18)$$

$$> S1$$

$$\frac{100 (10 + xd1)^2}{10.7500 + xd1 + 0.7500 xd2} \quad (19)$$

$$> P1$$

$$\frac{100 (10 + xd1)^2}{10.7500 + xd1 + 0.7500 xd2} - xd1 \left(1 + \frac{xd1}{2} \right) \quad (20)$$

$$> S2$$

$$\frac{100 (1 + xd2)^2}{8.5000 + 0.7500 xd1 + xd2} \quad (21)$$

$$> P2$$

$$\frac{100 (1 + xd2)^2}{8.5000 + 0.7500 xd1 + xd2} - xd2 \left(1 + \frac{xd2}{2} \right) \quad (22)$$

$$> dP1d := \text{diff}(P1, xd1)$$

$$dP1d := \frac{200 (10 + xd1)}{10.7500 + xd1 + 0.7500 xd2} - \frac{100 (10 + xd1)^2}{(10.7500 + xd1 + 0.7500 xd2)^2} - 1 - xd1 \quad (23)$$

> #Determine the effort in reverse-engineering of firm 1

not assigned(xe1);

if not assigned(xe1) **then**

dP1e := diff(P1, xe1);

solve({dP1e=0, dP1d=0}, { xe1, xd1 }) :

[seq(s, s in %)] :

select(sol_set → is(0 < eval(xe1, sol_set)), %) :

xe1 := piecewise(eval(xe1, %[1]) < 0, 0, eval(xe1, %[1]) > (Q02 - Q01), (Q02 - Q01),
eval(xe1, %[1]));

end if

false (24)

> #Find best response of firm 1

solve(dP1d=0, xd1) :

> br1 := select(expr → **not** has(expr, I), %)

$$br1 := 0.0833 \left(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \right. \quad (25)$$

$$\begin{aligned} &+ 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \\ &- 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \left. \right)^{1/2} \Big)^{1/3} - \left(12.0000 \left(-0.0625 \, xd2^2 - 18.2917 \, xd2 \right. \right. \\ &- 1338.3403 \left. \right) \Big) / \left(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \right. \\ &+ 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \\ &- 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \left. \right)^{1/2} \Big)^{1/3} - 0.5000 \, xd2 + 25.8333 \end{aligned}$$

$$> dP2d := diff(P2, xd2)$$

$$dP2d := \frac{200 (1 + xd2)}{8.5000 + 0.7500 \, xd1 + xd2} - \frac{100 (1 + xd2)^2}{(8.5000 + 0.7500 \, xd1 + xd2)^2} - 1 - xd2 \quad (26)$$

$$> dP1d;$$

$$dP2d;$$

$$solve(\{dP1d=0\}, \{xd1\});$$

$$\begin{aligned} &\frac{200 (10 + xd1)}{10.7500 + xd1 + 0.7500 \, xd2} - \frac{100 (10 + xd1)^2}{(10.7500 + xd1 + 0.7500 \, xd2)^2} - 1 - xd1 \\ &\frac{200 (1 + xd2)}{8.5000 + 0.7500 \, xd1 + xd2} - \frac{100 (1 + xd2)^2}{(8.5000 + 0.7500 \, xd1 + xd2)^2} - 1 - xd2 \end{aligned}$$

$$\left\{ xd1 = 0.0833 \left(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \right. \quad (27)$$

$$\begin{aligned} &+ 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \\ &- 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \left. \right)^{1/2} \Big)^{1/3} - \left(12.0000 \left(-0.0625 \, xd2^2 - 18.2917 \, xd2 \right. \right. \\ &- 1338.3403 \left. \right) \Big) / \left(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \right. \\ &+ 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \\ &- 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \left. \right)^{1/2} \Big)^{1/3} - 0.5000 \, xd2 + 25.8333 \Big\}, \left\{ xd1 = \right. \end{aligned}$$

$$-0.0417 \left(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \right.$$

$$\begin{aligned} &+ 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \\ &- 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \left. \right)^{1/2} \Big)^{1/3} + \left(6.0000 \left(-0.0625 \, xd2^2 - 18.2917 \, xd2 \right. \right. \end{aligned}$$

$$\begin{aligned}
& -1338.3403) \Big) \Big/ \Big(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \\
& + 180.0000 \, (-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \\
& - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8)^{1/2} \Big)^{1/3} - 0.5000 \, xd2 + 25.8333 \\
& + 0.8660 \, \text{I} \Bigg(0.0833 \, \Big(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \\
& + 180.0000 \, (-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \\
& - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8)^{1/2} \Big)^{1/3} + (12.0000 \, (-0.0625 \, xd2^2 - 18.2917 \, xd2 \\
& - 1338.3403) \Big) \Big/ \Big(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \\
& + 180.0000 \, (-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \\
& - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8)^{1/2} \Big)^{1/3} \Bigg) \Bigg\} , \Bigg\{ xd1 = -0.0417 \, \Big(27.0000 \, xd2^3 \\
& - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \\
& + 180.0000 \, (-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \\
& - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8)^{1/2} \Big)^{1/3} + (6.0000 \, (-0.0625 \, xd2^2 - 18.2917 \, xd2 \\
& - 1338.3403) \Big) \Big/ \Big(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \\
& + 180.0000 \, (-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \\
& - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8)^{1/2} \Big)^{1/3} - 0.5000 \, xd2 + 25.8333 \\
& - 0.8660 \, \text{I} \Bigg(0.0833 \, \Big(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7
\end{aligned}$$

$$\begin{aligned}
& + 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \\
& \left. - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \right)^{1/2} \bigg)^{1/3} + \left(12.0000 \left(-0.0625 \, xd2^2 - 18.2917 \, xd2 \right. \right. \\
& \left. \left. - 1338.3403 \right) \right) / \left(27.0000 \, xd2^3 - 36747.0000 \, xd2^2 + 1.6373 \, 10^6 \, xd2 + 8.4556 \, 10^7 \right. \\
& \left. + 180.0000 \left(-81.0000 \, xd2^5 + 37179.0000 \, xd2^4 - 4.9831 \, 10^6 \, xd2^3 - 2.6382 \, 10^8 \, xd2^2 \right. \right. \\
& \left. \left. - 5.1254 \, 10^8 \, xd2 - 2.5374 \, 10^8 \right)^{1/2} \bigg)^{1/3} \bigg) \}
\end{aligned}$$

> #Determine the effort in reverse-engineering of firm 2

not assigned(xe2);

if not assigned(xe2) **then**

$dP2e := \text{diff}(P2, xe2);$

$\text{solve}(\{dP2e=0, dP2d=0\}, \{xe2, xd2\}) :$

$[\text{seq}(s, s \text{ in } \%)] :$

$\text{select}(\text{sol_set} \rightarrow \text{is}(0 < \text{eval}(xe2, \text{sol_set})), \%):$

$xe2 := \text{piecewise}(\text{eval}(xe2, \%[1]) < 0, 0, \text{eval}(xe2, \%[1]) > (Q01 - Q02), (Q01 - Q02),$
 $\text{eval}(xe2, \%[1]));$

end if

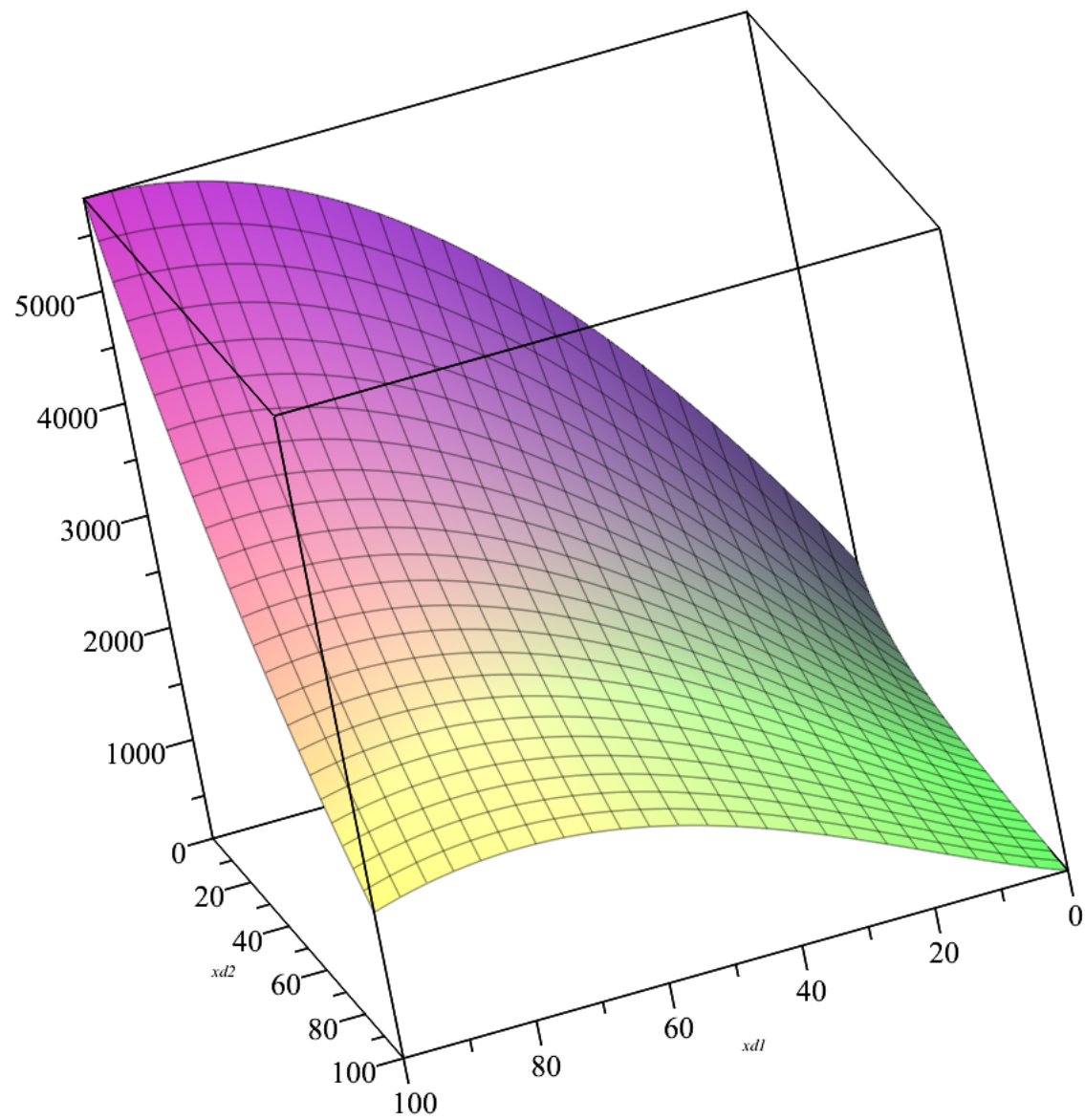
false

(28)

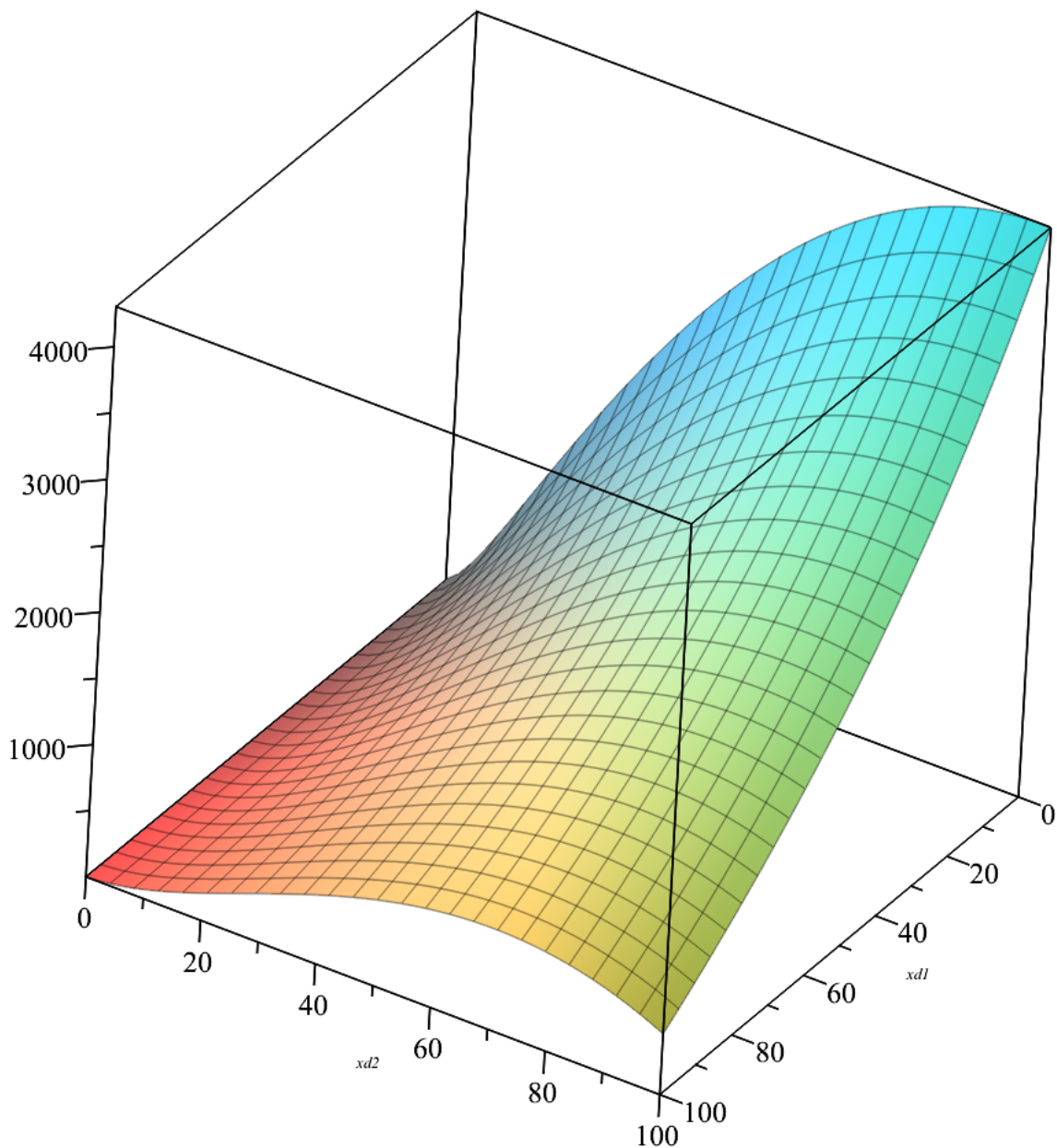
>

> #Payoff function for firm 1

$\text{plot3d}(P1, xd1 = 0 .. 100, xd2 = 0 .. 100)$



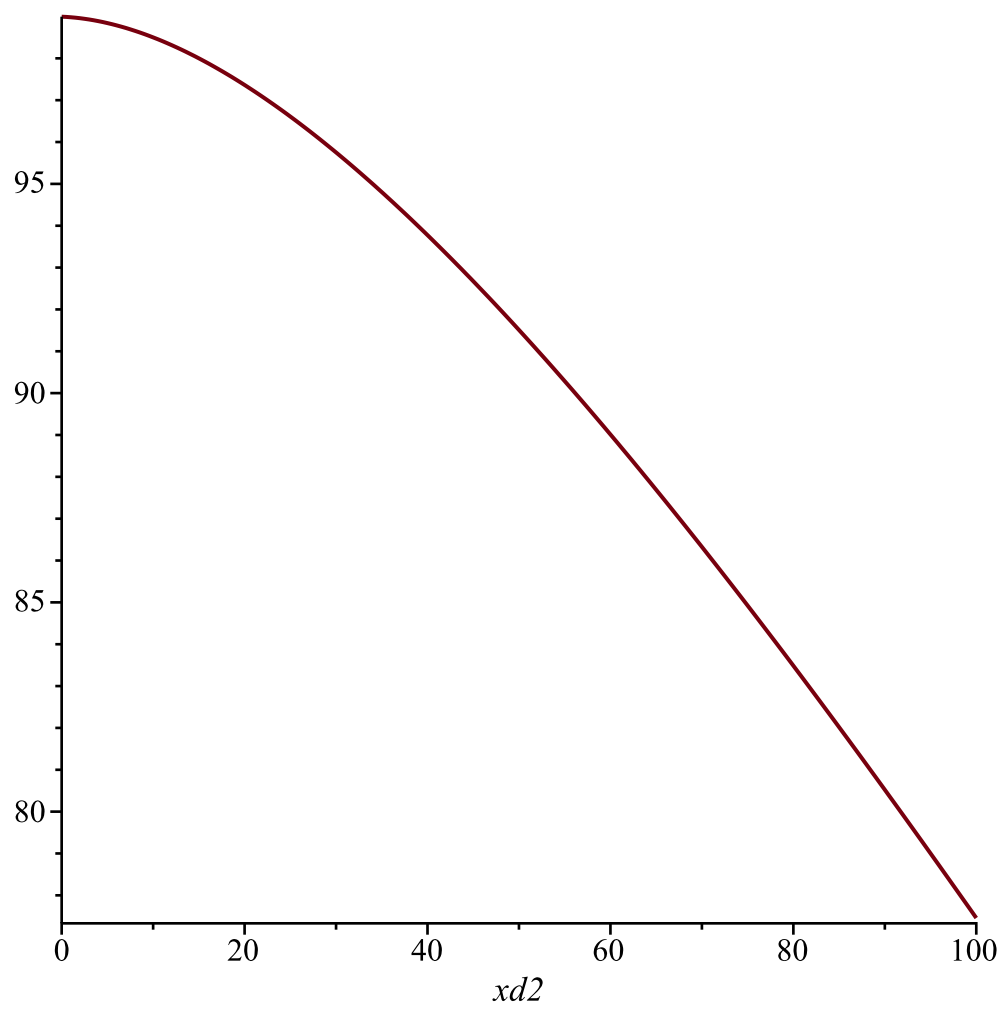
```
> #Payoff function for firm 2  
plot3d(P2, xd1 = 0 ..100, xd2 = 0 ..100)
```



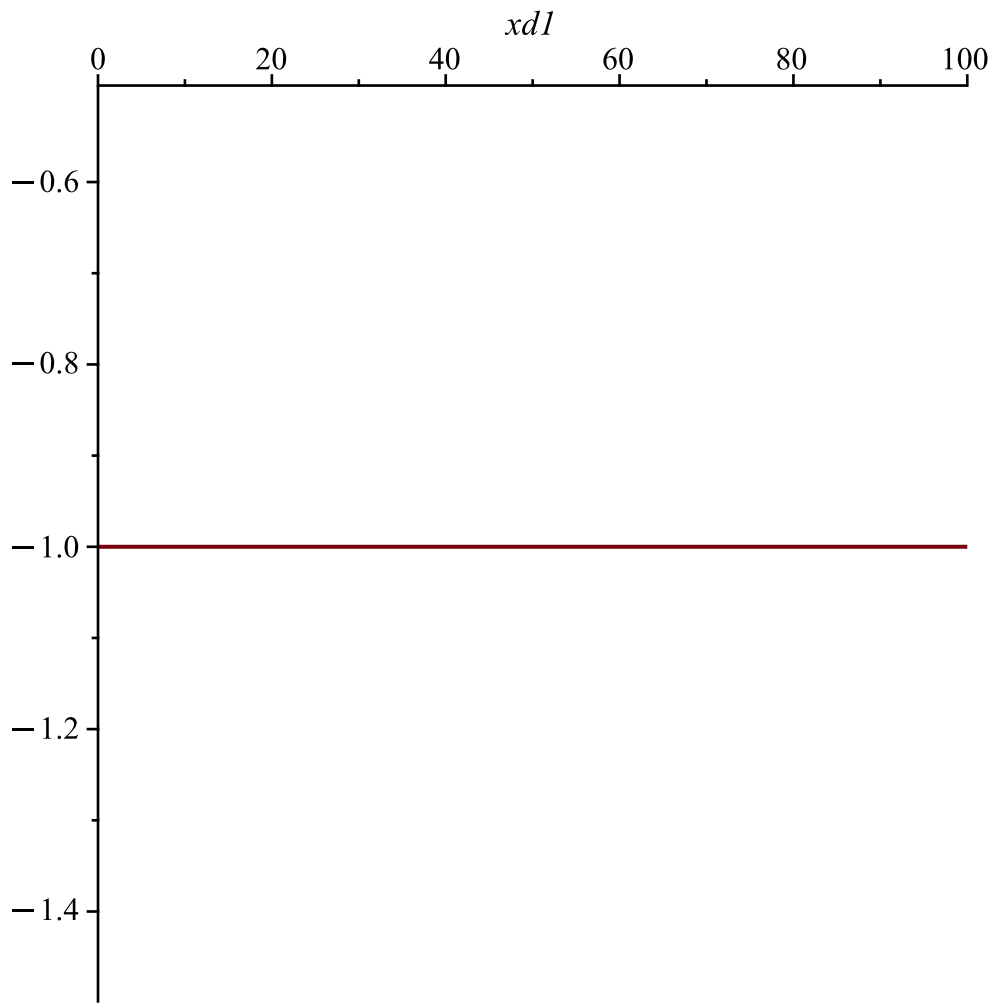
```

> #Find best response of firm 2
  solve(dP2d=0, xd2)
-1.0000, -0.7500 xd1 + 41.5000 + 5.0000  $\sqrt{3.0000 xd1 + 130.0000}$ , -0.7500 xd1 + 41.5000 (29)
  - 5.0000  $\sqrt{3.0000 xd1 + 130.0000}$ 
> br2 := select(expr → not has(expr, I), %)
  br2 := -1.0000 (30)
>
> #Best response of xd1 for firm 1
  plot(br1, xd2=0..100, )

```



```
> #Best response of xd2 for firm 2  
plot(br2, xd1 = 0..100, )
```



```

>
> #Find equilibrium
> solve( {dP1d=0, dP2d=0}, {xd1, xd2})
{xd1 = 99.0000, xd2 = -1.0000}, {xd1 = 84.6417, xd2 = 75.9888}, {xd1 = -62.3282
+ 624.7369 I, xd2 = -62.5058 - 623.9579 I}, {xd1 = -14.6383, xd2 = 98.8698}, {xd1
= -62.3282 - 624.7369 I, xd2 = -62.5058 + 623.9579 I}
(31)
> solution_list := [seq(s, s in %) ] :
> CheckRealSolutionStructure := proc(sol_struct)
    local list_of_eqs;
    list_of_eqs := convert(sol_struct, list);
    andmap(eq → is(rhs(eq), real), list_of_eqs);
end proc:
> select(CheckRealSolutionStructure, solution_list) :
> out := evalf(allvalues(%))
out := [ {xd1 = 99.0000, xd2 = -1.0000}, {xd1 = 84.6417, xd2 = 75.9888}, {xd1 = -14.6383,
xd2 = 98.8698} ]
(32)
> #[seq(elem, elem in %) ] :
> for inner_list in out do
    listsol := [op(listsol), op(inlist)];

```

end do;

Error, recursive assignment

```
> select( listsol → is(0 < eval(xd1, listsol)) and is(0 < eval(xd2, listsol)), %) [1]
                                     {xd1 = 84.6417, xd2 = 75.9888} (33)
```

```
> assign(%)
```

```
>
```

```
> xe1
```

0 (34)

```
> xe2
```

0 (35)

```
> xd1
```

84.6417 (36)

```
> xd2
```

75.9888 (37)

```
> Q1
```

94.6417 (38)

```
> Q2
```

76.9888 (39)

```
> P1
```

2211.2254 (40)

```
> P2
```

1042.5876 (41)

```
>
```

```
>
```